

Higher order non-homogeneous LDE with constant coefficients.

(Method of Undetermined coefficients) (UC Method)

UC Function:

1. x^n

2. $\sin ax$

3. $\cos ax$

4. e^{ax}

UC set:

$\{x^n, x^{n-1}, x^{n-2}, \dots, x^2, x, 1\}$

$\{\sin ax, \cos ax\}$

$\{\cos ax, \sin ax\}$

$\{e^{ax}\}$

* An equation of the following form

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots$$

$+ a_n y = F(x)$
is called non-homogeneous LDE with constant coefficients.

OR $a_0 y^{(n)} + a_1 y^{(n-1)} + a_2 y^{(n-2)} + \dots + a_n y = F(x)$

$\therefore 4y'' - 6y' + 9y = \sin x + x^2 + e^{2x}$

UC set: $\{x^2, x, 1\}$

$\therefore x^2, 2x, 1$

$\therefore e^{2x} \{e^{2x}\}$

$\therefore \sin x = \{\sin x, \cos x\}$

* Find the general solution of each of the following differential equation.

1) $y'' - 2y' - 3y = 2e^x - 10\sin x, y(0) = 2, y'(0) = 4$

2) $y'' - 4y' + 3y = 9x^2 + 4, y(0) = 6, y'(0) = 3$

3) $y'' - 4y' + 13y = 8\sin 3x$

4) $y'' - 2y' - 8y = 4e^{2x} - 21e^{-3x}$

5) $y'' - 4y' + 13y = 18e^{-2x}$

Higher order non-homogeneous LDE with constant coefficients:

Step ①

→ Given eqn → ①

→ L.H.S = 0

Let, $y = e^{mx}$

$$y' = me^{mx}$$

$$y'' = m^2 e^{mx}$$

⋮

→ put value in eqn ②

$$(\quad) e^{mx} = 0$$

$$e^{mx} \neq 0$$

$m =$ values

∴ Distinct: $m = \alpha, \beta, \gamma$

$$eq^n \Rightarrow y_1 = c_1 e^{\alpha x} + c_2 e^{\beta x} + c_3 e^{\gamma x}$$

∴ Equal: $m = \alpha, \alpha, \alpha$

$$eq^n \Rightarrow y_1 = c_1 e^{\alpha x} + c_2 x e^{\alpha x} + c_3 x^2 e^{\alpha x}$$

∴ Complex: $m = \alpha \pm i\beta$

$$eq^n \Rightarrow y_1 = e^{\alpha x} (c_1 \sin \beta x + c_2 \cos \beta x)$$

$$y_1 = \text{---}$$

Step ②

R.H.S

$$\rightarrow UC: x^n \rightarrow \{x^n, x^{n-1}, x^{n-2}\}$$

$$: \sin ax \rightarrow \{\sin ax, \cos ax\}$$

$$: \cos ax \rightarrow \{\cos ax, \sin ax\}$$

$$: e^{ax} \rightarrow \{e^{ax}\}$$

→ do Union to find S

→ put values in eqn ①

$$y_2 = A B C$$

$$y_2' =$$

$$y_2'' =$$

→ simplify

Compare L.H.S with R.H.S and take coefficients

→ Find A, B, C

$$\rightarrow y_2 = \text{---} \text{ from UC set}$$

Step-3

$$\text{General sol}^n \text{ is } y = y_1 + y_2$$

$$y = \text{---}$$

IF y & y' given

find c_1 & c_2

(Ans)

$$1) y'' - 2y' - 3y = 2e^x - 10\sin x \rightarrow (1)$$

$$\therefore y'' - 2y' - 3y = 0 \rightarrow (2)$$

Let, $y = e^{mx}$ be trial of solⁿ of eqn (2)

$$y' = me^{mx}$$

$$y'' = m^2 e^{mx}$$

putting values in eqn (2)

$$\therefore m^2 e^{mx} - 2me^{mx} - 3e^{mx} = 0$$

$$\Rightarrow (m^2 - 2m - 3)e^{mx} = 0$$

$\therefore$ Since $e^{mx} \neq 0$, auxiliary eqn will be,

$$\Rightarrow m^2 - 2m - 3 = 0$$

$$\Rightarrow m^2 - 3m + m - 3 = 0$$

$$\Rightarrow (m+1)(m-3) = 0$$

$$\Rightarrow m = -1, 3 \text{ [distinct]}$$

$$y_1 = c_1 e^{-x} + c_2 e^{3x}$$

Again from (1),

$$s_1 = \{e^x\}, s_2 = \{\sin x, \cos x\}$$

(Union)

UC set,

$$s = \{e^x, \sin x, \cos x\}$$

$$y_2 = Ae^x + B\sin x + C\cos x$$

$$y_2' = Ae^x + B\cos x - C\sin x$$

$$y_2'' = Ae^x - B\sin x - C\cos x$$

put y_2, y_2', y_2'' in (1)

$$Ae^x - B\sin x - C\cos x -$$

$$2Ae^x - 2B\cos x + 2C\sin x -$$

$$3Ae^x - 3B\sin x - 3C\cos x$$

$$= 2e^x - 10\sin x$$

$$\Rightarrow (A - 2A - 3A)e^x + (-B + 2C - 3B)\sin x + (-C + 2C - 3C)\cos x = 2e^x - 10\sin x$$

$$\Rightarrow -4Ae^x.$$

$$Ae^x - B\sin x - C\cos x$$

$$-2Ae^x - 2B\cos x + 2C\sin x$$

$$-3Ae^x - 3B\sin x - 3C\cos x$$

$$= 2e^x - 10\sin x$$

$$\Rightarrow e^x(A - 2A - 3A) + \sin x(-B + 2C - 3B) + \cos x(-C - 2B - 3C) = 2e^x - 10\sin x$$

Comparing,

$$\therefore A - 2A - 3A = 2 \quad -B + 2C - 3B = -10$$

$$\Rightarrow -4A = 2 \quad \Rightarrow -4B + 2C = -10$$

$$\Rightarrow A = -\frac{1}{2} \quad \Rightarrow -2B + C = -5$$

$$-C - 2B - 3C = 0 \quad \Rightarrow -2(-2C) + C = -5$$

$$\Rightarrow 4C + C = -5$$

$$\Rightarrow 5C = -5$$

$$\Rightarrow -4C = 2B$$

$$\Rightarrow -2C = B$$

$$\Rightarrow B = 2$$

$$y_2 = -\frac{1}{2}e^x - 2\sin x + 2\cos x$$

$$y_2 = -\frac{1}{2}e^x + 2\sin x \cos x$$

General eqⁿ:

$$y = y_1 + y_2$$

$$\Rightarrow y = c_1 e^{-x} + c_2 e^{3x} - \frac{1}{2} e^x + 2 \sin x + \cos x$$

given, $y(0) = 2$
 $y'(0) = 4$

$$\Rightarrow 2 = c_1 e^0 + c_2 e^0 - \frac{1}{2} e^0 + 2(0) + 1$$

$$2 = c_1 + c_2 - \frac{1}{2} \quad |$$

~~2 = 1~~

$$\Rightarrow \cancel{c_1 + c_2 = 1} \Rightarrow c_1 + c_2 = \frac{5}{2}$$

Σ_1

$$y' = -c_1 e^{-x} + 3c_2 e^{3x} - \frac{1}{2} e^x + 2 \cos x + \sin x$$

$$\Rightarrow 4 = -c_1 e^0 + 3c_2 e^0 - \frac{1}{2} e^0 + 2(1) + 0$$

$$\Rightarrow 4 = -c_1 + 3c_2 - \frac{1}{2} + 2$$

$$\Rightarrow c_1 - 3c_2 = \cancel{-6} - \frac{5}{2}$$

$$c_1 = 2, \quad c_2 = \frac{3}{2}$$

$$y = 2e^{-x} + \frac{3}{2}e^{3x} - \frac{1}{2}e^x + 2\sin x + \cos x$$

$$2) y'' - 4y' + 3y = 9x^2 + 4, y(0) = 6, y'(0) = 3$$

L.H.S:

$$y'' - 4y' + 3y = 0 \rightarrow \textcircled{1}$$

Let, $y = e^{mx}$

$$y' = me^{mx}$$

$$y'' = m^2 e^{mx}$$

$$\Rightarrow m^2 e^{mx} - 4me^{mx} + 3e^{mx} = 0$$

$$\Rightarrow (m^2 - 4m + 3)e^{mx} = 0$$

$\therefore$ Since $e^{mx} \neq 0$, auxiliary eqn will be

$$\Rightarrow m^2 - 4m + 3 = 0$$

$$\Rightarrow m^2 - 3m - m + 3 = 0$$

$$\Rightarrow m(m-3) - 1(m-3) = 0$$

$$\Rightarrow (m-3)(m-1) = 0$$

$$\Rightarrow m = 3, 1 \text{ [distinct]}$$

$$y_1 = C_1 e^{3x} + C_2 e^x$$

R.H.S:

$$S = \{x^2, x, 1\}$$

$$y_2 = Ax^2 + Bx + C$$

$$\Rightarrow y_2' = 2Ax + B$$

$$\Rightarrow y_2'' = 2A$$

$$\Rightarrow 2A - 8Ax + 4B + 3Ax^2 + 3Bx + 3C = 9x^2 + 4$$

$$\Rightarrow 3Ax^2 + (-8A + 3B)x + (2A - 4B + 3C) = 9x^2 + 4$$

Comparing,

$$3A = 9$$

$$A = 3$$

$$-8A + 3B = 0$$

$$\Rightarrow -8 \times 3 + 3B = 0$$

$$\Rightarrow 3B = 24$$

$$\Rightarrow B = 8$$

$$2A - 4B + 3C = 4$$

$$\Rightarrow 6 - 32 + 3C = 0$$

$$\Rightarrow 3C = 26$$

$$\Rightarrow C = 10$$

$$y_2 = 3x^2 + 8x + 10$$

General eqn,

$$y = y_1 + y_2$$

$$\Rightarrow y = C_1 e^{3x} + C_2 e^x + 3x^2 + 8x + 10$$

$$\Rightarrow y' = 3C_1 e^{3x} + C_2 e^x + 6x + 8$$

$$\Rightarrow y(0) = 3C_1 + C_2 + 0 + 8 = 6$$

$$\Rightarrow 3C_1 + C_2 = -2$$

$$\Rightarrow 6 = C_1 + C_2 + 10$$

$$\Rightarrow C_1 + C_2 = -4$$

$$C_1 = -\frac{1}{2}, C_2 = -\frac{7}{2}$$

$$\Rightarrow y = -\frac{1}{2} e^{3x} - \frac{7}{2} e^x + 3x^2 + 8x + 10$$

(A)

$$(3) y'' - 4y' + 13y = 8 \sin 3x \rightarrow (1)$$

L.H.S:

$$y'' - 4y' + 13y = 0 \rightarrow (2)$$

$$\text{Let } y = e^{mx}$$

$$y' = me^{mx}$$

$$y'' = m^2 e^{mx}$$

$$\Rightarrow m^2 e^{mx} - 4me^{mx} + 13e^{mx} = 0$$

$$\Rightarrow (m^2 - 4m + 13)e^{mx} = 0$$

Since $e^{mx} \neq 0$, eqn will be:

$$\Rightarrow m^2 - 4m + 13 = 0$$

$$\Rightarrow m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{4 \pm \sqrt{16 - 52}}{2}$$

$$= \frac{4 \pm \sqrt{-36}}{2} \Rightarrow \frac{4 \pm \sqrt{36} \cdot \sqrt{-1}}{2}$$

$$= \frac{2 \pm \sqrt{-36}}{1} \Rightarrow \frac{2 \pm 6i}{1}$$

$$= 2 \pm \sqrt{36} \cdot \sqrt{-1} \Rightarrow 2(2 \pm 3i)$$

$$\Rightarrow 2 \pm 6i \text{ [complex]} \Rightarrow 2 \pm 3i \text{ [complex]}$$

$$\Rightarrow y_1 = e^{2x} (c_1 \sin 3x + c_2 \cos 3x)$$

$$S = \{ \sin 3x, \cos 3x \}$$

R.H.S:

$$y_2 = A \sin 3x + B \cos 3x \rightarrow (3)$$

$$\Rightarrow y_2' = 3A \cos 3x - 3B \sin 3x$$

$$\Rightarrow y_2'' = -9A \sin 3x - 9B \cos 3x$$

$$-9A \sin 3x - 9B \cos 3x$$

$$-12A \cos 3x + 12B \sin 3x$$

$$+ 13A \sin 3x + 13B \cos 3x = 8 \sin 3x$$

$$\Rightarrow (-9A + 12B + 13A) \sin 3x + (-9B - 12A + 13B) \cos 3x = 8 \sin 3x$$

$$\Rightarrow (4A + 12B) \sin 3x + (4B - 12A) \cos 3x = 8 \sin 3x$$

Comparing:

$$\Rightarrow 4A + 12B = 8$$

$$\Rightarrow 4A +$$

$$\Rightarrow A + 3B = 2$$

$$\Rightarrow A + 3(3A) = 2$$

$$\Rightarrow 10A = 2$$

$$\Rightarrow A = \frac{1}{5}$$

$$4B - 12A = 0$$

$$\Rightarrow 4B = 12A$$

$$\Rightarrow B = 3A$$

$$\Rightarrow B = \frac{3}{5}$$

$$y_2 = \frac{1}{5} \sin 3x + \frac{3}{5} \cos 3x$$

General eqn:

$$y = y_1 + y_2$$

$$\Rightarrow y = e^{2x} (c_1 \sin 3x + c_2 \cos 3x) + \frac{1}{5} \sin 3x + \frac{3}{5} \cos 3x$$

(C)

(Ans)

$$y'' - 2y' - 8y = 4e^{2x} - 21e^{-3x} \quad \rightarrow \textcircled{1}$$

L.H.S:

$$y'' - 2y' - 8y = 0 \quad \textcircled{2}$$

Let, $y = e^{mx}$

$$y' = me^{mx}$$

$$y'' = m^2 e^{mx}$$

$$m^2 e^{mx} - 2me^{mx} - 8e^{mx} = 0$$

$$(m^2 - 2m - 8)e^{mx} = 0$$

Since, $e^{mx} \neq 0$ eqⁿ will be

$$m^2 - 2m - 8 = 0$$

$$m^2 - 4m + 2m - 8 = 0$$

$$m(m - 4) + 2(m - 4) = 0$$

$$(m + 2)(m - 4) = 0$$

$$m = -2, 4 \text{ [distinct]}$$

$$y_1 = C_1 e^{-2x} + C_2 e^{4x}$$

R.H.S:

$$S_1 = \{e^{2x}\}, S_2 = \{e^{-3x}\}$$

UC set,

$$S = \{e^{2x}, e^{-3x}\}$$

$$y_2 = Ae^{2x} + Be^{-3x} \quad \rightarrow \textcircled{3}$$

$$y_2' = 2Ae^{2x} - 3Be^{-3x}$$

$$y_2'' = 4Ae^{2x} + 9Be^{-3x}$$

$$4Ae^{2x} + 9Be^{-3x}$$

$$-2Ae^{2x} + 6Be^{-3x}$$

$$-8Ae^{2x} - 8Be^{-3x} = 4e^{2x} - 21e^{-3x}$$

$$(4A - 4A - 8A)e^{2x} + (9B + 6B - 8B)e^{-3x} =$$

$$-8Ae^{2x} + 7Be^{-3x} = 4e^{2x} - 21e^{-3x}$$

Comparing:

$$-8A = 4 \quad | \quad 7B = -21$$

$$A = -\frac{1}{2} \quad | \quad B = -3$$

$$y_2 = -\frac{1}{2}e^{2x} - 3e^{-3x}$$

General eqn:

$$y = y_1 + y_2$$

$$y = C_1 e^{-2x} + C_2 e^{4x} - \frac{1}{2}e^{2x} - 3e^{-3x}$$

$$(5) \quad y'' - 4y' + 13y = 18e^{-2x}$$

L.H.S: $\rightarrow$ ①

$$y'' - 4y' + 13y = 0 \rightarrow$$

$$\text{Let, } y = e^{mx}$$

$$y' = me^{mx}$$

$$y'' = m^2 e^{mx}$$

$$\therefore m^2 e^{mx} - 4me^{mx} + 13e^{mx} = 0$$

$$\Rightarrow (m^2 - 4m + 13) e^{mx} = 0$$

Since $e^{mx} \neq 0$, eqn will be:

$$\therefore m^2 - 4m + 13 = 0$$

$$\Rightarrow m = \frac{4 \pm \sqrt{-36}}{2}$$

$$m = 2 \pm 3i \text{ [complex]}$$

$$y_1 = e^{2x} (C_1 e^{3x} + C_2 \sin 3x + C_3 \cos 3x)$$

R.H.S:

$$S = \{e^{-2x}\}$$

$$\Rightarrow y_2 = Ae^{-2x}$$

$$\Rightarrow y_2' = -2Ae^{-2x}$$

$$\Rightarrow y_2'' = 4Ae^{-2x}$$

$$\Rightarrow 4Ae^{-2x} + 8Ae^{-2x} + 13Ae^{-2x} = 18e^{-2x}$$

$$\Rightarrow 25Ae^{-2x} = 18e^{-2x}$$

Comparing

$$25A = 18$$

$$A = 18/25$$

$$\Rightarrow y_2 = \frac{18}{25} e^{-2x}$$

General

$$y = y_1 + y_2$$

$$\Rightarrow y = e^{2x} \{C_1 \sin(3x) + C_2 \cos(3x)\} + \frac{18}{25} e^{-2x}$$